The Variance Risk Premium over Trading and Non-Trading Periods

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In this paper we decompose the variance risk premium into overnight and intraday components using model-free implied variance stock indices in US, Europe and Asia. We find that during the non-trading overnight period the variance risk premium is significantly negative, whereas during the intraday trading period the variance risk premium becomes positive and often insignificant. We also assess the predictive ability of the overnight and intraday variance risk premiums with respect to future equity returns. We find that the intraday component performs better at shorter prediction horizons whereas the overnight variance risk premium performs better at longer horizons. Our empirical results suggest that non-trading effects are an important determinant of the variance risk premium.

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1. Introduction

It is well known in the literature that average delta-hedged option returns are negative (see, for example, Bakshi and Kapadia, 2003) because the volatility embedded in option prices tends to be higher than the realized volatility (see, for example Bollerslev et al., 2009). The negative risk premium embedded in option prices indicates that investors are willing to pay an insurance premium to buy protection against unexpected changes in volatility. Recent studies have uncovered an interesting phenomenon related to the different behavior of option returns during trading and non-trading periods. Muravyev and Ni (2020) find that daily delta-hedged overnight option returns in the S&P 500 are typically negative, whereas intraday option returns turn positive. Similarly, Bhat et al. (2024) find that Nifty options display negative overnight returns and positive intraday returns.

Motivated by these intriguing findings, in this paper we decompose the variance risk premium into its overnight and intraday components. More specifically, we use equity model-free implied variance indices in the US and abroad and decompose the variance risk premium into its overnight and intraday components using the daily mark-to-market returns of a hypothetical portfolio of a variance swap. Variance swaps provide pure exposure to the return variance of the underlying asset and hence the return of a variance swap quantifies the variance risk premium better than delta hedged option returns. Then we use predictive regressions to test the predictive ability of the variance risk premium components with respect to future equity returns. To the best of our knowledge, our study is the first to decompose the variance risk premium into overnight and intraday components using a variance swap approach and test the predictive ability of the overnight and intraday components of the variance risk premium.

In the extant literature nmany studies use the difference between the model-free implied variance and realized variance in order to quantify the variance risk premium. Although it is a well-documented fact that variance carries a negative risk premium, the underlying factors that generate this premium are not well understood. The variance risk premium may reflect compensation for bearing the risk of unexpected changes in volatility (see for example, e.g., Bali & Zhou, (2016); Bollerslev & Todorov, (2011); Carr & Wu, (2009); Drechsler & Yaron, (2011)), it may be generated by market frictions or the presence of financial intermediaries in the option market (see for example, e.g., Chan et al., (1995); Fournier & Jacobs, (2020); Gârleanu et al., (2009); Muravyev, (2016); Santa-Clara & Saretto, (2009)), or it may be the result of mispricing and sentiment volatility (see for example, e.g., Bondarenko, (2014); Branger & Schlag, (2008); Goyal & Saretto, (2009); Han, (2008); Jones & Shemesh, (2018); Muravyev & Ni, (2020); Poteshman, (2001); Weinbaum et al., (2023)). Given the prevalence of the variance risk premium in asset pricing studies, the contribution of this paper is to provide some further insights into the nature of the variance risk premium by decomposing the daily variance risk premium into overnight and intraday components using model-free implied variance indices.

Model-free implied variance represents the risk-neutral expected value of the return variance and is constructed using the prices of out-of-the-money calls and puts (see, for example, e.g., Carr and Madan, (1998); Demeterfi et al., (1999); Britten-Jones and Neuberger, (2000); Jiang and Tian, (2005); Carr and Wu, (2009), Dotsis and Vlastakis, (2016)). Many studies in the literature show that the difference between realized variance and model-free implied variance is, on average, negative, and hence, variance swap buyers are willing to pay a premium to hedge unexpected changes in volatility.

In the empirical analysis, we use the daily mark-to-market profit and loss of a hypothetical variance swap to decompose the daily (close-to-close) variance swap return into its overnight (close-to-open) and intraday (open-to-close) components. We use model-free implied variance indices based on major stock indices in the US market (S&P 500, Nasdaq-100, and Russell 2000) and some major stock indices outside the US (Europe and Asia). In the empirical analysis we also employ implied variance indices on individual stocks. We find that the pricing of variance risk is significantly different during the overnight and intraday periods. In particular, we find that the overnight variance swap returns are persistently negative and statistically significant, while intraday variance swap returns are mostly positive and sometimes insignificant. We observe the same pattern of overnight versus intraday variance returns when we calculate the variance swap returns using implied variance indices in US, Europe and Asia or stock implied variance indices.

We also estimate monthly predictive regressions of stock indices' returns using the variance risk premiums as the predictor variable. In line with previous studies in the literature (e.g., Bollerslev et al., 2009) we find that in the case of the S&P 500 the variance risk premium is a significant predictor of stock returns at short-term horizons. When we decompose the variance risk premium into its two components, we find that the intraday variance risk premium has better predictive ability over short horizons whereas the overnight variance risk premium performs better at longer horizons. These results are confirmed using both panel and individual predictive regressions.

The remainder of the paper proceeds as follows. The next Section presents a literature review. Section 3 describes the methodology for constructing variance swap returns. Section 4 discusses the data and presents the empirical results and Section 5 concludes.

2. Literature review

The different behavior of stock markets during the intraday period and the non-trading period is a well-documented fact and many explanations have been suggested in the extant literature. For example, Cliff et al. (2008) argue that the high opening prices are partly responsible for the positive overnight returns. Kelly & Clark (2011) consider whether this could be a result of the behavior by some traders that liquidate their positions towards the end of the session and reestablish them at the next opening or the result of not fully professional traders who feel increased confidence over their position during trading hours. Lou et al. (2019) argue that the heterogeneity of investors and the different participation of them during different periods of the day could be responsible for the persistence of the overnight and intraday different behavior of returns. As institutional investors and arbitrageurs avoid initiating trades at the opening of the session, this leaves space for the uniformed investors to define with their trades the return sign and the magnitude of the opening. As the initial momentum subsides, and the participation of arbitrageurs and institutional investors increases, the behavior of returns changes. Bogousslavsky (2021) argues that institutional constraints, margin parameters, and price pressure caused by the reluctance of arbitrageurs to trade at the opening of the session, while they are eager to reduce their positions near market closing, supports the different behavior in non-trading versus intraday returns. The phenomenon of overnight versus intraday return is also documented in futures markets. The electronically traded (mini) S&P futures contracts are practically traded 24 hours a day (with a short time interruption). However, their returns do not accrue linearly during the trading session (e.g., Boyarchenko et al., 2021). Almost all of the average market return accrues around the opening of European markets (Bondarenko and Muravyev, 2023).

The difference between the intraday and non-trading period returns is also documented in option markets. Delta hedged option returns are on average negative (Bakshi & Kapadia, 2003) as the implied variance tends to be higher than the realized one (Bollerslev and Zhou, 2006). However, average options returns are not distributed evenly around the clock. It is well documented that options display different performance among different periods of the day. Sheikh and Ronn (1994) find that option returns are more negative during non-trading periods (mainly during weekends) and positive towards the end of the trading session. Jones and Shemesh (2018) confirm that during non-trading periods and especially on weekends, options present lower returns. As options present different behavior during trading and non-trading hours, so do their implied variances. While the underlying's realized variance is decreased substantially during non-trading hours, implied variance seems to incorporate the expectation that variance will remain sizable even when markets are closed.

Muravyev and Ni (2020) find that while (delta hedged) option returns are on average negative, overnight returns are on average more negative, while intraday returns become on average positive. They argue that for this puzzling change of signs, the most promising explanation is the "volatility seasonality" bias where market participants are acting "as if" the day and night instantaneous volatilities are the same while, as previously documented, the per-hour volatility is higher during trading hours compared to non-trading hours. Also implied variance seems to behave differently during trading hours and non-trading session. Chen et al. (2021) decompose daily changes of VIX and find different average behavior of VIX during intraday and overnight periods. They find that intraday and overnight VIX changes are negative correlated, and they argue that the unavailability of trading during the non-trading hours creates uncertainty while trading helps to resolve uncertainty.

Wang et al. (2022) argue that returns and volatility outside trading hours contain unique information rather than just being supplementary data to intraday data. Building on this insight, they develop a model that distinguishes between intraday returns and overnight returns and show that their model reduces significantly option pricing errors. Liang et al. (2023) also develop an option pricing model based on different intraday and overnight variance dynamics. They show that this model offers improved accuracy in pricing SPX options particularly during periods of high volatility.

Motivated by these findings, we examine if the intraday versus overnight behavior is observed in variance swaps and if this behavior is similar in other markets and indices besides the S&P 500. In the empirical analysis, we decompose the variance risk premium into its overnight and intraday components using the mark-to-market returns of a hypothetical variance swap position with a maturity of one calendar month (21 trading days). In our paper, we find that the returns of variance swaps are different during the intraday and the non-trading periods not only in the S&P 500 but also in other markets. The overnight variance swap return tends to be negative, while the intraday return tends to be positive.

3. P&L of Variance Swaps

A variance swap is a contract that provides direct exposure to the variance of an underlying asset (*S*) for a predefined time period. We calculate the daily close-to-close profit and loss (P&L) of the variance swap as follows:

$$P\&L_{t,close-to-close} = D\left[\begin{array}{l}\frac{d}{30}\left(\frac{360}{d}\times\left(100\cdot ln\left(\frac{S_{t,close}}{S_{t-1,close}}\right)\right)^2 - IV_{t-1,close}^2\right)\\ +\frac{30-d}{30}\left(IV_{t,close}^2 - IV_{t-1,close}^2\right)\end{array}\right] \quad (1)$$

According to the above formulae, the daily P&L of the variance swap is equal to the realized payout (realized variance minus initial strike price) weighted by the elapsed time (*d*) and the change in the variance swap rate between time *t*-1 and *t*. This is the P&L that an investor would accrue by opening a long position in a variance swap at time *t-1* and closing the position by going short the variance swap at time *t*. In the ensuing empirical analysis, we approximate the variance swap rate using model-free implied variances (IV) with 30 days to maturity. Since we are using implied variances with constant time to maturity of 30 calendar days to maturity at both time *t-d* and time *t*, as in Daniel and Moskowitz (2016), we implicitly make the assumption that the short end of the variance swap term structure is flat, i.e., the swap rate with a maturity of 30 calendar days (21 trading days) is identical to the swap rate with a maturity of 29 calendar days (20 trading days). We are using calendar days in the computation of the P&L, and the factor *d* in the above formula takes the value of three (3) on Mondays and the value of one (1), on any other day of the week. This is crucial in order not to overestimate the realized variance in relation to the implied one during weekends.

We decompose the daily close-to-close P&L of the variance swap into close-to-open and open-to-close as follows:

$$P\&L_{t,close-to-open} = D\left[\begin{array}{l}\frac{d-tr\%}{30}\left(\frac{360}{d-tr\%}\times\left(100\cdot ln\left(\frac{S_{t,open}}{S_{t-1,close}}\right)\right)^2 - IV_{t-1,close}^2\right)\\ +\frac{(30-d+tr\%)}{30}\left(IV_{t,open}^2 - IV_{t-1,close}^2\right)\end{array}\right] \quad (2)$$

$$P\&L_{t,open-to-close} = P\&L_{t,close-to-close} - P\&L_{t,close-to-open} \quad (3)$$

The factor *tr*% represents the percentage of the day between opening and closing and takes into account the trading hours per underlying. Most markets are open for 6.5 hours. However European markets are open for 8.5 hours and Nikkei for 6 hours. Thus, the factor *tr*% for the majority of markets is equal to 6.5/24=0.27083, for European markets is equal to 0.35417 and for Nikkei it is equal to 0.25. Since we are interested in the difference in the performance of the variance swap P&L during the intraday and the non-trading period, the inclusion of a discount factor (*D*) should produce a negligible difference in the results and thus we omit it in the calculations.

4. Data and Discussion of Empirical Results

4.1 Data Descriptive Statistics

The data used in the empirical analysis consists of model-free implied volatility indices with 30-day maturity. From the available model-free implied volatility indices we choose those that have available close and open prices for both the indices and the underlying assets for a sufficient period of time. The implied volatility indices that satisfy our criteria are presented in Table 1. The data were downloaded from Bloomberg and the common sample period is from 30/01/2012 until 30/09/2022. The implied volatility indices used in the empirical analysis have, as underlying assets, major stock indices in the US market (S&P 500, Nasdaq-100 and Russell 2000) and some major stock indices outside US (Europe, Japan, Germany, Korea, Switzerland). In the empirical analysis we also employ implied volatility indices on stocks (Apple, Goldman Sachs, Google, IBM, and Amazon).

[Table 1]

We begin our analysis by investigating the behavior of implied volatility changes on close-to-open and the open-to-close periods. The results are reported in Table 2. As in Chen et al., 2021 we find that changes in volatility indices are on average positive during the close-to-open (non-trading) period and negative during the open-to-close (trading) period. The negative average changes of implied volatility during trading hours and the average positive changes during non-trading hours is an indication that trading helps to resolve the uncertainty that has accumulated during the non-trading period (see also Chen et al., 2021).

The results are statistically significant in all cases. For the close-to-open (Co) period from the 14 implied volatility indices twelve are statistically significant at 1% level and two at the 5% significance level. For the open-to-close (oC) period, eleven results are statistically significant at 1% level and three at 5% significance level. The results regarding their difference (Co-oC) are all statistically significant at the 1% level. Since the same results are observed across different markets worldwide, it reinforces the idea that the observed relation is associated with the different behavior of implied volatility indices (and option markets) during trading and nontrading hours and it is not due to VIX or U.S. market specific characteristic.

[Table 2]

4.2 Daily P&L of the variance swaps

In Table 3 we present for each underlying the average P&L of the variance swap during the whole trading day, the nontrading overnight period (close-to-open) and the intraday period (open-to-close). The first column presents the average daily close-to-close P&L of the variance swap.

The P&L is negative for all cases under consideration, though statistically significant in six underlyings. The second column in Table 3 shows that during nontrading hours, the P&L of the

variance swap (P&LCo) results in losses in all underling assets under consideration, and these losses are statistically significant in almost all cases. In contrast, the average intraday P&L of the variance swap (P&LoC) is positive in ten out of the fourteen underling assets and the point estimates are statistically significant in six cases. Since the average daily close-to-close P&L of the variance swap is negative it follows that the close-to-open P&L dominates, and it is the most significant component of the total daily P&L in variance swaps.

[Table 3]

The results so far suggest that, in line with previous results in the literature, the variance risk premium, as proxied by the variance swap return, is negative. The decomposition of the variance swap return shows that the negative variance risk premium can be attributed to the negative premium that accrues during the nontrading period. Thus, a strategy of staying long at a variance swap during trading hours and reversing that position to short during the nontrading period would generate a positive P&L. The last column confirms that intuition as this strategy presents statistically significant profits (before transaction costs) in eleven out of the fourteen underling assets.

4.2.1. Day of the Week effect

Next, we examine the possible existence of day-of-the-week effects in variance swaps returns. More specifically, we test for day-of-the-week effects using the following regression:

$$P\&L_t = \phi_M M_t + \phi_T T_t + \phi_W W_t + \phi_{TH} TH_t + \phi_F F_t + \varepsilon_t \qquad (4)$$

where M_t, T_t, W_t, TH_t, F_t are the dummy variables for each day of the week, and $\varphi's$ are their respective coefficients. We omit the constant term and apply the Newey & West (1987) adjustment

to address possible autocorrelation or heteroscedasticity. Table 4 presents the results for the close-to-close, the close-to-open and the open-to-close returns, respectively.

[Table 4]

Table 4 shows that variance swap returns are mainly negative. In fact, all statistically significant coefficients are negative. The Monday P&L is negative in all cases, and statistically significant in six out of the fourteen underlying assets. We also observe that the Monday negative P&L, in all the statistically significant cases except SMI, is the most negative P&L of the week. As the P&L of variance swaps is based on the difference between realized variance and previous period implied variance, it is reasonable that on Mondays this difference to present its lowest value. On the closing of Friday typically option prices and their respective volatilities increase to compensate option writers for the possibility of new significant information announcements during the non-trading period of the weekend. A notable result is that on Fridays the only significant results are the negative P&L's of the three US indices and of one US individual stock. This is an indication that realized variance in US remains low (compared to the implied one) ahead of the non-trading period of the weekend.

The close-to-open (non-trading hours) P&L return presents a clearer pattern as almost all results are negative and statistically significant. On Monday in twelve cases the returns are significant and on Tuesday and Wednesday that number is reduced to nine and ten, respectively. At the last two days of the week, while all the statistically significant results are also negative, most results are insignificant. Monday's close-to-open P&L is the weekend's P&L from Friday's closing until the opening of Monday. As the implied variance at the end of the week is usually overpriced to address the increase in the uncertainty of the long non-trading period of the weekend,

and as the per hour realized variance decreases during non-trading hours, it is reasonable for the weekend close-to-open P&L to be among the most negative P&L of all the non-trading periods.

Finally, we observe a varying behavior of the intraday P&L (P&LoC), depending on the day of the week. The statistically significant results on Mondays are negative, while on Tuesdays they are mixed, with three negative and two positive. However, during the rest days of the week, except one case, all the statistically significant P&LoC are positive.

4.2.2. P&L on Expiration Days

Next we examine the average variance swaps P&Ls at option expiration dates (third Friday of every month or if that day is a holiday the previous trading day). More specifically, we test if the variance swap P&L at expiration days is different from the average P&L during the corresponding month. For each calendar month, we subtract from the expiration day P&L the average P&L of the month and then we take the average of these differences. Table 5 reports the results.

[Table 5]

The P&L on expiration days is not significantly different from the monthly average P&L, except for two cases where the P&L on expiration days is more negative than the monthly average. We conclude that that daily variance swap returns do not present significantly different behavior on expiration days. The same also applies for the case of the overnight period and the intraday period. Overall, the results presented in Table 5 suggest that on expiration days the overnight and intraday variance risk premiums do not display significantly different behavior compared to all other days.

4.2.3. The effect of Variance-of-Variance

In this section we test if the behavior of variance swap returns varies with the level of variance-of-variance. Currently there are two indices of implied volatility of implied volatility available, the VVix (implied volatility of VIX) and VVstoxx (the implied volatility of VSTOXX). We hypothesize that during periods of high implied volatility of implied volatility, the variance swap returns, and variance risk premiums may become more negative.

First, we split into quartiles the values of VVix and VVstoxx. Then, per quartile, we determine the dates that these values occurred, and for these dates, we calculate the next day's P&L of the variance swap and decompose it into close-to-close, close-to-open, and open-to-close. We examine if higher values of vol-of-vol indices are leading to a higher or lower P&L of the variance swap of the corresponding index over the next day. The results are presented in Table 6.

[Table 6]

The results presented in Table 6 suggest that higher values of vol-of-vol indices are related to significantly more negative overnight P&Ls. More specifically, both in the case of VVix and VVstoxx the overnight P&L decreases almost monotonically from the 1st quartile to 4th quartile. The intraday P&L in most cases is not statistically significant across the different quartiles. These results suggest that the level uncertainty regarding variance risk is an important determinant of the overnight variance risk premium.

5. Variance Risk Premium and Stock Return Predictability

This section examines the predictive ability of the overnight and intraday components of the variance risk premium using stock indices (S&P 500, NASDAQ 100, RUSELL 2000, SMI, DAX, EURO STOXX 50, NIKKEI 225, KOSPI and HSI).

First, we use the following regression for each asset separately to test the predictive ability of the variance risk premiums.:

$$R_{t,t+h} = a_h + \beta_h VRP_t + \varepsilon_{t,t+h} \tag{5}$$

where $R_{t,t+h}$ is the cumulative return of the underlying asset h months ahead, and VRP is either the total variance risk premium, the overnight variance risk premium, or the intraday variance risk premium as calculated from the respective variance swap P&Ls. We use forecasting horizons of 1, 3, 6, 9, and 12 months, respectively. We use monthly data to estimate regression (5). We calculate the monthly variance risk premiums (total, overnight, intraday) by summing their daily counterparts over a month.[3] The monthly time series data (October 2007 to September 2022) of the total, overnight and intraday variance risk premiums of the S&P 500 are plotted in Figure 1. In line with the previously discussed results, Figure 1 shows that the monthly overnight variance risk premium is more negative than the monthly intraday variance risk premium.

[Insert Figure 1]

Table 7 reports the slope estimates from regression (5) when the predictive variable is the monthly variance risk premium (Panel A), the overnight monthly variance risk premium (Panel B) and the monthly intraday variance risk premium (Panel C). Similar to previous studies in the literature (e.g., Bollerslev et al., 2009) the results reported in Table 7 suggest that the variance risk premium is a good predictor of equity returns at forecasting horizons between 3-9 months. For

[3] We also include a dummy variable to take into account the significant spike in March of 2020. Note that without the inclusion of the dummy the VRP loses its predictive ability with respect to S&P 500 returns.

example, at the 3-month horizon the VRP attains an R^2 of about 5% in the case of S&P 500, Nasdaq100 and Russell2000, respectively. The negative slope coefficient suggests that as the variance risk premium becomes more negative the expected equity returns increase. In the case of stock indices outside US the slope coefficients become significant mainly at the 6-month and 9-month horizon, respectively. Note that the in the case of NIKKEI 225 and HSI the coefficient of the variance risk premium is insignificant at all forecasting horizons.

Panel B and Panel C report the slope coefficients when we decompose the variance risk premium into its overnight and intraday components, respectively. The slope coefficient of the overnight variance risk premium is significant at the 9–12-month horizon. When the predictor variable is the intraday variance risk premium the slope coefficient is statistically significant at the 1–6-month horizon. For example, in the case of the S&P 500 the slope coefficient is negative and statistically significant, and the predictive regression attains an R^2 of about 6.1% at the one-month forecasting horizon.

[Table 7]

We also analyze the predictability of VRP measures using panel regressions as in Bollerslev et al. (2014) and Qiao et al. (2024). We perform panel regressions with country-level fixed effects and estimate the following model:

$$R_{t,t+h}^{i} = a_h + \beta_h VRP_t^i + \gamma^i + \varepsilon_{t,t+h}^{i} \tag{6}$$

where $R_{t,t+h}^{i}$ is the cumulative return of the underlying asset h months ahead, and γ^i captures the country fixed effects. As previously, the forecasting horizon (h) is 1, 3, 6, 9, and 12 months. Table

8 presents the results from the panel regressions. The close-to-close variance risk premium measure (VRP), displays negative coefficients that are statistically significant at all prediction horizons. The intraday variance risk premium (open-to-close) displays significant coefficients at the 1 to 3 month forecasting horizons, while the overnight variance risk premium (close-to-open) displays significant coefficients at the 6 to 12 month forecasting horizons. The slope coeffcients suggest that a more negative total, overnight and intraday variance risk premium predicts higher expected returns.

[Table 8]

Previous studies in the literature argue that the variance risk premium is a proxy for representative agent's risk aversion or alternatively a proxy for macroeconomic uncertainty (see, for example, Zhou (2018) and the references therein). Our empirical results suggest that the total variance risk premium contains mixed information. The intraday variance risk premium component appears to capture short-term risks and performs better at shorter prediction horizons. Conversely, the non-trading period component of the variance risk premium serves as a better return prediction variable for longer horizons and seems to contain different information, likely reflecting longer-term risks.

6. Conclusion

This paper uses the P&L of a hypothetical variance swap position and breaks down the variance risk premium into its overnight and intraday components to provide some further insights on the nature of the variance risk premium. In the empirical analysis we use implied variance stock indices in US, Europe and Asia and find that the variance risk premium switches sign between overnight and intraday periods. During the overnight period the variance risk premium is negative

and during the intraday trading period the variance risk premium becomes positive. Our findings suggest that the negative variance risk premium reported in numerous previous studies is primarily driven by the overnight period variance risk premium component. We also evaluate the ability of the intraday and overnight variance risk premium to predict future equity returns. We find that that the intraday component captures short-term risk and displays predictive ability at 1–3-month horizons, while the overnight component reflects longer term risk and displays predictive ability at 6-12-month horizons.

A promising avenue for future research is to incorporate overnight variance risk into equilibrium models of consumption risk and examine whether these models can successfully capture key characteristics of asset returns and option prices.

Table 1: List of Implied Volatility Indices

Name	Name	Underlying	Country
Cboe Volatility Index	VIX	S&P 500	US
Cboe NASDAQ Volatility Index	VXN	Nasdaq-100	US
Cboe Russell 2000 Vol. Index	RVX	Russell	US
Volatility Index on the SMI	VSMI	SMI	CH
Deutsche Börse's Volatility index	VDAX-NEW	DAX	GER
Euro Stoxx 50 Volatility	VSTOXX	Euro Stoxx	EUREX
Nikkei Stock Average Volatility	VNK	Nikkei 225	JP
KOSPI 200 Volatility Index	VKOSPI	KOSPI 200	KR
HSI Volatility Index	VHSI	Hang Seng	HK
Cboe Apple VIX Index	VXAPL	AAPL	US
Cboe Goldman Sachs VIX Index	VXGS	GS	US
Cboe Google VIX Index	VXGOG	GOOGL	US
Cboe IBM VIX Index	VXIBM	IBM	US
Cboe Amazon VIX Index	VXAZN	Amazon	US

Table 2: Daily changes in implied volatility indices during overnight (Co) and intraday (oC) periods. Results are in volatility points. The symbols *, **, *** represent 10%, 5% and 1% significance levels. The time period is from 30/01/2012 until 30/09/2022.

Volatility Index	Close-Open (Co) Avg	Open-Close (oC) Avg	Difference Co-oC
VIX	0,12968*** (7,0391)	-0,12498*** (-4,0863)	0,25465*** (6,8432)
VXN	0,10250*** (8,2378)	-0,09629*** (-3,1813)	0,19879*** (5,9459)
RVX	0,18959*** (10,5486)	-0,18502*** (-6,4975)	0,37461*** (10,7127)
VSMI	0,20654*** (25,5435)	-0,20486*** (-8,8704)	0,41140*** (16,3737)
VDAX-NEW	0,13043*** (8,0539)	-0,12783*** (-5,4610)	0,25827*** (9,6248)
VSTOXX	0,05647** (2,4952)	-0,05420** (-2,2432)	0,11067*** (3,3248)
VNK	0,10244*** (4,4060)	-0,10016*** (-4,1648)	0,20260*** (5,5599)
VKOSPI	0,13211*** (8,1621)	-0,13091*** (-7,6251)	0,26302*** (11,1235)
VHSI	0,04153** (2,1015)	-0,03778** (-2,0397)	0,07931*** (2,9383)
VXAPL	0,10867*** (3,3327)	-0,09894** (-2,3686)	0,20761*** (3,4720)
VXGS	0,28460*** (7,8473)	-0,28108*** (-6,2881)	0,56568*** (8,1608)
VXGOG	0,13744*** (4,8576)	-0,12867*** (-4,0957)	0,26611*** (6,0266)
VXIBM	0,15889*** (4,4152)	-0,15143*** (-3,6889)	0,31033*** (4,7616)
VXAZN	0,17225*** (5,1969)	-0,16829*** (-5,0728)	0,34054*** (7,3580)

Table 3: The profit and loss (P&L) of a variance swap position during overnight (Co) and intraday (oC) periods. The symbols *, **, *** represent 10%, 5%, and 1% significance levels. The time period is from 30/01/2012 until 30/09/2022.

Volatility Index	**P&L**	**P&L$_{Co}$**	**PLoC**	**PL$_{DIF}$**
VIX	-3,25443	-6,14867***	2,89424	9.043***
	(-1,0748)	(-5,3512)	(1,0684)	(3.1677)
VXN	-2,62075	-6,21973***	3,59898	9.819***
	(-0,8867)	(-6,6586)	(1,3455)	(3.6297)
RVX	-4,11915	-10,63584***	6,51670*	17.153***
	(-1,1331)	(-8,0204)	(1,9489)	(4.8201)
VSMI	-3,21765*	-0,91405*	-2,30360	-1.390
	(-1,8879)	(-1,6797)	(-1,3719)	(-0.7619)
VDAX-NEW	-4,34330*	-4,16286***	-0,18044	3.982*
	(-1,6692)	(-3,2469)	(-0,0870)	(1.7584)
VSTOXX	-6,91652**	-13,97410***	7,05758***	21.032***
	(-2,4961)	(-8,5744)	(3,3491)	(8.2393)
VNK	-4,93561**	-8,90354***	3,96793**	12.871***
	(-2,2224)	(-6,6369)	(2,1253)	(5.4205)
VKOSPI	-2,53063*	-1,63990*	-0,89073	0.749
	(-1,7513)	(-1,7429)	(-0,7319)	(0.4606)
VHSI	-3,78124*	-7,70507***	3,92382***	11.629***
	(-1,9200)	(-5,5879)	(2,9012)	(6.1439)
VXAPL	-6,71074	-16,80028***	10,08954**	26.890***
	(-1,4382)	(-5,7417)	(2,4299)	(4.9231)
VXGS	-6,68555	-5,40499	-1,28056	4.124
	(-1,2003)	(-1,5181)	(-0,2741)	(0.6693)
VXGOG	-3,08747	-7,80135***	4,71388*	12.515***
	(-0,8691)	(-3,8056)	(1,7064)	(3.7654)
VXIBM	-3,04359	-6,60994**	3,56635	10.176*
	(-0,7549)	(-2,0971)	(0,9799)	(1.8547)
VXAZN	-6,29979	-10,57247***	4,27268	14.845***
	(-1,5554)	(-4,3598)	(1,2043)	(3.2763)

Table 4: The profit of a variance swap position (close-to-close, open-to-close, close-to-open) during different days of the week. The symbols *, **, *** represent 10%, 5% and 1% significance levels. The time period is from 01/02/2012 until 30/09/2022.

Panel A: Close-to-Close					
Underlying	**MON**	**TUE**	**WED**	**THU**	**FRI**
SP500	-2,92417	-4,37525	-4,07387	5,80071	-10,68596**
	(-0,2684)	(-1,2910)	(-1,0753)	(0,6934)	(-2,2681)
NASDAQ100	-6,23861	-2,41696	-1,82773	5,62354	-8,20733*
	(-0,5771)	(-0,6512)	(-0,4504)	(0,7576)	(-1,8995)
RUSSELL2000	-11,64370	-4,73642	3,01875	5,34424	-12,60004**
	(-0,8900)	(-1,0964)	(0,4144)	(0,5747)	(-2,2386)
SMI	-10,02729***	-10,73081**	-2,04298	3,58297	3,06285
	(-2,8961)	(-2,1144)	(-1,0522)	(0,8527)	(0,6940)
DAX	-14,70466**	-9,30439**	-4,20051	4,91366	1,46281
	(-2,1748)	(-2,1680)	(-1,2187)	(0,6411)	(0,3085)
EUROSTOXX50	-18,58675***	-11,14737**	-6,30934	1,03293	0,31749
	(-2,8460)	(-2,1677)	(-1,4697)	(0,1280)	(0,0603)
NIKKEI225	-26,36796***	-9,61716*	-0,20030	5,81260	5,67545
	(-5,2204)	(-1,8436)	(-0,0499)	(1,0624)	(1,0976)
KOSPI	-4,79238	-9,62413***	0,44257	-0,72707	2,15341
	(-1,2368)	(-3,9213)	(0,1711)	(-0,1671)	(0,6701)
HSI	-6,08654	-13,24108**	1,13649	-0,73017	-0,08115
	(-1,0666)	(-2,2142)	(0,3122)	(-0,2356)	(-0,0224)
AAPL	-21,15063	-12,45544*	-7,55241	12,34629	-4,73194
	(-1,4023)	(-1,6484)	(-0,9172)	(1,1209)	(-0,5900)
GS	-31,70480*	-16,39398	13,31202	5,27908	-4,07213
	(-1,8177)	(-1,5466)	(0,8975)	(0,3646)	(-0,4769)
GOOGL	-12,15366	-4,55631	1,97300	6,77278	-7,45718
	(-1,1543)	(-0,7650)	(0,3082)	(0,8356)	(-0,8883)
IBM	-34,82831***	10,63665	6,90865	9,33258	-7,30298
	(-4,2965)	(1,0897)	(0,7420)	(0,8888)	(-0,9410)
AMZN	-14,84010	-3,26001	-6,78171	15,23947	-23,01361**
	(-1,5397)	(-0,4833)	(-0,9303)	(1,5868)	(-2,1541)
Panel B: Close-to-Open					
Underlying	**MON**	**TUE**	**WED**	**THU**	**FRI**
SP500	-7,54974***	-9,09192***	-8,82027***	-2,03612	-3,29466
	(-2,8523)	(-4,1215)	(-3,9762)	(-0,6706)	(-1,2616)
NASDAQ100	-18,33913***	-3,94377**	-4,13491***	-2,28400	-2,49953
	(-6,3186)	(-2,1277)	(-3,1414)	(-1,2481)	(-1,1191)
RUSSELL2000	-19,31965***	-11,39398***	-7,68075**	-6,99527***	-7,86581***
	(-6,1191)	(-3,7269)	(-2,0462)	(-3,0269)	(-2,9626)
SMI	-2,65686**	-2,06750	-1,02495	-0,50677	1,66916

	(-2,4092)	(-1,0208)	(-1,6304)	(-0,4974)	(1,3606)
DAX	-4,92051	-8,37258***	-5,65295***	-2,58493	0,65682
	(-1,2425)	(-4,5512)	(-3,4411)	(-1,2108)	(0,1717)
EUROSTOXX50	-16,16376***	-19,51004***	-11,13927***	-10,84315***	-12,24612***
	(-3,7882)	(-4,6316)	(-3,8593)	(-4,4739)	(-2,6009)
NIKKEI225	-18,69733***	-10,15335***	-6,01149***	-5,40216*	-4,29599
	(-5,4318)	(-2,9814)	(-2,7374)	(-1,9319)	(-1,3539)
KOSPI	-2,42168	-3,86383**	-1,41466	-4,25960**	3,81481
	(-1,0898)	(-2,0911)	(-0,9747)	(-2,2166)	(1,2082)
HSI	-9,24237***	-11,78941***	-6,92697***	-5,75356**	-4,68668
	(-2,8035)	(-2,9201)	(-3,0869)	(-2,4084)	(-1,3705)
AAPL	-32,59181***	-11,39100	-16,03222***	-6,66172	-17,39368***
	(-4,7169)	(-1,5280)	(-2,9415)	(-0,8270)	(-3,1158)
GS	-14,76720**	-1,15875	-0,30247	-10,36149	-0,43419
	(-1,9874)	(-0,1290)	(-0,0385)	(-1,3159)	(-0,0751)
GOOGL	-10,27076*	-3,80077	-11,33123***	-5,79572	-7,80045
	(-1,6845)	(-0,8254)	(-3,0950)	(-1,3993)	(-1,5854)
IBM	-21,78000***	-1,71394	-3,97610	-9,63164	3,98767
	(-4,0977)	(-0,2886)	(-0,5993)	(-0,9934)	(0,6085)
AMZN	-14,09072***	-7,92155*	-17,85667***	-6,01887	-6,78563
	(-2,7203)	(-1,9387)	(-5,3347)	(-1,4894)	(-0,7582)

Panel C: Open-to-Close

Underlying	**MON**	**TUE**	**WED**	**THU**	**FRI**
SP500	4,62556	4,71667	4,74640	7,83683	-7,39130**
	(0,4577)	(1,5259)	(1,1771)	(1,1890)	(-2,1124)
NASDAQ100	12,10052	1,52681	2,30719	7,90754	-5,70781
	(1,2415)	(0,4742)	(0,5903)	(1,2859)	(-1,2388)
RUSSELL2000	7,67596	6,65756	10,69950**	12,33951	-4,73423
	(0,6124)	(1,6176)	(2,2928)	(1,4551)	(-0,9378)
SMI	-7,37044**	-8,66331**	-1,01803	4,08974	1,39369
	(-2,2300)	(-2,0292)	(-0,5274)	(0,9579)	(0,3016)
DAX	-9,78415**	-0,93181	1,45244	7,49859	0,80599
	(-2,1198)	(-0,2762)	(0,5003)	(1,2012)	(0,1960)
EUROSTOXX50	-2,42299	8,36267***	4,82993*	11,87608*	12,56360***
	(-0,5021)	(3,0286)	(1,7264)	(1,7514)	(2,9490)
NIKKEI225	-7,67063**	0,53619	5,81119	11,21476**	9,97143**
	(-2,0505)	(0,1370)	(1,5715)	(2,2096)	(2,2015)
KOSPI	-2,37070	-5,76031***	1,85723	3,53254	-1,66140
	(-0,8414)	(-2,7066)	(0,7229)	(0,9184)	(-0,6276)
HSI	3,15583	-1,45167	8,06346**	5,02339**	4,60553*
	(0,9086)	(-0,4455)	(2,4574)	(2,1320)	(1,7124)
AAPL	11,44118	-1,06444	8,47981	19,00801**	12,66173*
	(0,8095)	(-0,1486)	(1,1145)	(2,3075)	(1,7740)

GS	-16,93760	-15,23523*	13,61449	15,64056	-3,63794
	(-1,1125)	(-1,9222)	(1,4535)	(1,2820)	(-0,4361)
GOOGL	-1,88290	-0,75554	13,30423**	12,56849**	0,34327
	(-0,2526)	(-0,1647)	(2,5053)	(1,9632)	(0,0565)
IBM	-13,04831**	12,35059*	10,88476	18,96423	-11,29065
	(-1,9826)	(1,9338)	(1,0339)	(1,5502)	(-1,4511)
AMZN	-0,74938	4,66154	11,07496*	21,25834***	-16,22798
	(-0,0996)	(0,7660)	(1,7600)	(2,7942)	(-1,6182)

Table 5: Profit of a variance swap position (close-to-close, open-to-close, close-to-open) during expiration days (third Friday of every month or if that day is a holiday the previous trading day). We test if the variance swap P&L at expiration days is different from the average P&L during the corresponding month. For each calendar month, we subtract from the expiration day P&L the average P&L of the month and then we take the average of these differences. The symbols *, **, *** represent 10%, 5% and 1% significance levels. The time period is from 30/01/2012 until 30/09/2022.

	Close-to-Close	**Close-to-Open**	**Open-to-Close**
Underlying	**Exp - M Av.**	**Exp - M Av.**	**Exp - M Av.**
SP500	-8,78921	-5,85309	-4,30423
	(-0,9208)	(-0,9288)	(-0,7764)
Nasdaq100	-9,81678	3,72995	-13,54673
	(-0,9566)	(0,8564)	(-1,5194)
Russell2000	-23,35647*	-2,35477	-21,00170**
	(-1,8493)	(-0,5485)	(-2,0879)
SMI	-3,76840	1,13096	-4,89937
	(-0,5853)	(0,8192)	(-0,8314)
DAX	-14,57555	-5,06944	-9,50611
	(-1,1241)	(-1,4087)	(-0,8771)
EUROSTOXX50	-14,72691	-16,44144	1,71452
	(-1,0266)	(-1,0900)	(0,2597)
NIKKEI225	3,88375	6,12007	-2,23632
	(0,6368)	(1,3828)	(-0,4375)
KOSPI	-4,32239	4,10553	-8,42792
	(-0,5823)	(1,3158)	(-1,0367)
HSI	-16,25337*	-9,85337	-6,40000
	(-1,7998)	(-0,8820)	(-1,1979)
AAPL	18,53951	5,31027	13,22924
	(0,8938)	(0,7153)	(0,6948)
GS	-22,15736	-3,80916	-18,34820
	(-1,0372)	(-0,4637)	(-0,8408)
GOOGL	3,47404	0,42083	3,05320
	(0,1348)	(0,0279)	(0,2136)
IBM	-37,61592	10,81219	-48,42811*
	(-1,3782)	(0,5511)	(-1,8371)
AMZN	-17,28687	-3,25798	-14,02889
	(-1,0138)	(-0,5136)	(-0,8949)

Table 6: Average return of variance swap positions (close-to-close, open-to-close, close-to-open) per quartile of the level of implied volatility of implied volatility index. The symbols *, **, *** represent 10%, 5% and 1% significance levels. The time period is from 23/03/2010 until 30/09/2022.

	VVix				**VVstoxx**			
quartile	1st	2nd	*3rd*	4th	1st	2nd	3rd	4th
Vol-of-Vol	78,77***	88,74***	98,67***	118,95***	69.33***	80.08***	90.28***	110.23***
P&L CC	-1,02	-4,2**	-0,84	-9,48	-4.72***	-6.32***	-1.65	-17.94**
	(-0,83)	(-2,55)	(-0,24)	(-0,90)	(-2.92)	(-3.00)	(-0.33)	(-2.01)
P&L oC	1,99*	-1,39	5,93**	8,39	2.84**	4.18**	18.02***	10.65
	(1,81)	(-0,92)	(2,16)	(0,90)	(2.16)	(2.23)	(4.33)	(1.57)
P&L Co	-3,01***	-2,82***	-6,77***	-17,88***	-7.56***	-10.5***	-19.67***	-28.59***
	(-4,90)	(-2,78)	(-3,88)	(-4,30)	(-8.98)	(-8.59)	(-7.64)	(-5.33)

Table 7: Predictive regressions using monthly total (close-to-close), overnight (close-to-open), intraday (open-to-close) variance risk premiums. The symbols *, **, *** represent 10%, 5% and 1% significance levels. The time period is from October 2007 until September 2022. In NIKKEI 225, KOSPI and HSI the time period is from February 2012 until September 2022.

Panel A: Close-to-Close

Underlying	1 month			3 months			6 months			9 months			1 Year		
	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2
SP500	0,00293	-0,00002***	0,0514	0,01239	-0,00003***	0,0490	0,03061*	-0,00003**	0,0207	0,05023**	-0,00003**	0,0076	0,07236***	-0,00002*	0,0006
	(0,8800)	(-2,6906)		(1,3094)	(-3,2577)		(1,7597)	(-2,1479)		(2,2522)	(-2,4715)		(2,8166)	(-1,7264)	
Nasdaq100	0,00645*	-0,00002***	0,0643	0,02398**	-0,00004***	0,0473	0,05750***	-0,00003*	0,0063	0,09120***	-0,00002	0,0002	0,12814***	-0,00001	-0,0040
	(1,6788)	(-3,4752)		(2,1609)	(-3,7575)		(2,8373)	(-1,6648)		(3,4786)	(-1,4439)		(4,2326)	(-0,7310)	
Russell2000	0,00138	-0,00002**	0,0535	0,00886	-0,00004***	0,0512	0,02705	-0,00003	0,0070	0,04369	-0,00002	-0,0020	0,06114**	-0,00001	-0,0049
	(0,3140)	(-2,4319)		(0,7484)	(-3,2059)		(1,2639)	(-1,2884)		(1,6424)	(-0,9862)		(1,9864)	(-0,4933)	
SMI	0,00085	0,00000	-0,0051	0,00260	-0,00001	0,0009	0,00785	-0,00002**	0,0121	0,01413	-0,00003***	0,0169	0,02230	-0,00003*	0,0073
	(0,2591)	(-0,2781)		(0,3048)	(-0,9291)		(0,5434)	(-2,0624)		(0,7700)	(-2,7787)		(1,0322)	(-1,8560)	
DAX	0,00139	-0,00001	0,0105	0,00790	-0,00001	-0,0011	0,02130	-0,00001	-0,0017	0,03406	-0,00003*	0,0038	0,05045*	-0,00002	-0,0015
	(0,3412)	(-1,6044)		(0,7087)	(-1,1555)		(1,1328)	(-1,2056)		(1,4445)	(-1,7798)		(1,8736)	(-1,1137)	
EUROSTOX X50	-0,00347	-0,00002	0,0200	-0,00424	-0,00002	0,0075	-0,00312	-0,00003**	0,0130	-0,00079	-0,00005**	0,0158	0,00837	-0,00003	0,0031
	(-0,7294)	(-1,5703)		(-0,3633)	(-1,5075)		(-0,1581)	(-1,9867)		(-0,0316)	(-2,5028)		(0,2918)	(-1,4799)	
NIKKEI225	0,00243	0,00000	-0,0027	0,00859	0,00000	-0,0047	0,02035	-0,00001	-0,0031	0,03448	0,00000	-0,0056	0,05055	0,00000	-0,0056
	(0,5273)	(-1,2501)		(0,6884)	(-0,8365)		(0,9759)	(-1,2069)		(1,2768)	(-0,0837)		(1,6177)	(0,1061)	
KOSPI	-0,00023	-0,00001***	0,0202	0,00247	-0,00001	0,0016	0,00909	-0,00001	-0,0025	0,01511	-0,00001	-0,0018	0,02504	-0,00001	-0,0051
	(-0,0600)	(-3,0239)		(0,2257)	(-1,1370)		(0,4728)	(-0,6165)		(0,5970)	(-0,5523)		(0,8597)	(-0,2327)	
HSI	-0,00445	0,00000	-0,0043	-0,00907	-0,00001	-0,0032	-0,01195	-0,00001	0,0001	-0,01473	0,00000	-0,0056	-0,01608	0,00000	-0,0056
	(-0,8483)	(-0,5470)		(-0,6807)	(-0,9492)		(-0,5410)	(-0,8193)		(-0,5186)	(-0,0448)		(-0,5053)	(0,0492)	

Panel B: Close-to-Open

Underlying	1 month			3 months			6 months			9 months			1 Year		
	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2	Constant	VS (Lag)	Adj R^2
SP500	0,00987***	0,00002*	0,0205	0,02061**	0,00002	0,0022	0,03502**	0,00000	-0,0056	0,04736**	-0,00003	-0,0014	0,06747**	-0,00003	-0,0008
	(2,8794)	(1,8005)		(2,2340)	(0,6349)		(2,0945)	(0,0766)		(2,0450)	(-0,8900)		(2,4720)	(-1,0781)	
Nasdaq100	0,00839**	0,00000	-0,0053	0,02764***	0,00000	-0,0056	0,05173***	-0,00005	0,0033	0,07535***	-0,00010**	0,0213	0,11261***	-0,00009*	0,0124
	(2,1369)	(-0,1561)		(2,8121)	(-0,0537)		(2,8502)	(-1,2422)		(2,8313)	(-2,3467)		(3,4104)	(-1,7012)	
Russell2000	0,00963**	0,00001	0,0106	0,01829	0,00001	-0,0020	0,02323	-0,00002	-0,0004	0,02482	-0,00006	0,0247	0,04127	-0,00006	0,0184
	(2,1584)	(1,2058)		(1,4771)	(0,3917)		(1,2231)	(-0,6098)		(0,9841)	(-1,5537)		(1,3738)	(-1,5111)	
SMI	0,00105	0,00000	-0,0055	0,00367	-0,00001	-0,0054	0,01120	-0,00007	0,0102	0,01922	-0,00013***	0,0309	0,02688	-0,00013*	0,0204
	(0,3353)	(-0,1835)		(0,4275)	(-0,1264)		(0,7899)	(-1,5933)		(1,0570)	(-2,6230)		(1,2689)	(-1,8709)	
DAX	0,00129	-0,00001	-0,0025	0,01023	0,00000	-0,0051	0,01522	-0,00004	0,0085	0,02660	-0,00005*	0,0131	0,03997	-0,00006**	0,0162
	(0,2680)	(-0,6155)		(0,9061)	(0,2037)		(0,7736)	(-1,5217)		(1,0855)	(-1,7484)		(1,4686)	(-1,9774)	
EUROSTOX X50	-0,00022	0,00000	-0,0056	0,00087	0,00000	-0,0052	-0,00531	-0,00002	0,0016	-0,01003	-0,00004**	0,0142	-0,00979	-0,00006***	0,0234
	(-0,0404)	(0,0698)		(0,0682)	(0,1548)		(-0,2496)	(-1,0994)		(-0,3720)	(-2,1023)		(-0,3190)	(-2,7969)	
NIKKEI225	0,00660	-0,00001	-0,0041	0,02390*	0,00000	-0,0079	0,04980**	-0,00001	-0,0076	0,07260***	-0,00004	0,0027	0,09855***	-0,00005	0,0029
	(1,1738)	(-0,5433)		(1,8940)	(-0,0319)		(2,5309)	(-0,2814)		(2,7600)	(-1,0690)		(3,2415)	(-1,2719)	
KOSPI	0,00006	-0,00002	-0,0044	0,00273	-0,00001	-0,0077	0,00770	-0,00003	-0,0063	0,01375	-0,00003	-0,0065	0,02535	0,00007	-0,0030
	(0,0144)	(-0,4370)		(0,2783)	(-0,1705)		(0,4387)	(-0,3647)		(0,5538)	(-0,3338)		(0,8099)	(0,5798)	
HSI	-0,00073	0,00002	-0,0027	-0,00179	0,00002	-0,0058	-0,00598	-0,00002	-0,0059	-0,00888	-0,00003	-0,0056	-0,01261	-0,00002	-0,0068
	(-0,1297)	(1,0708)		(-0,1509)	(0,7390)		(-0,2758)	(-0,5042)		(-0,3600)	(-0,7422)		(-0,4357)	(-0,5140)	

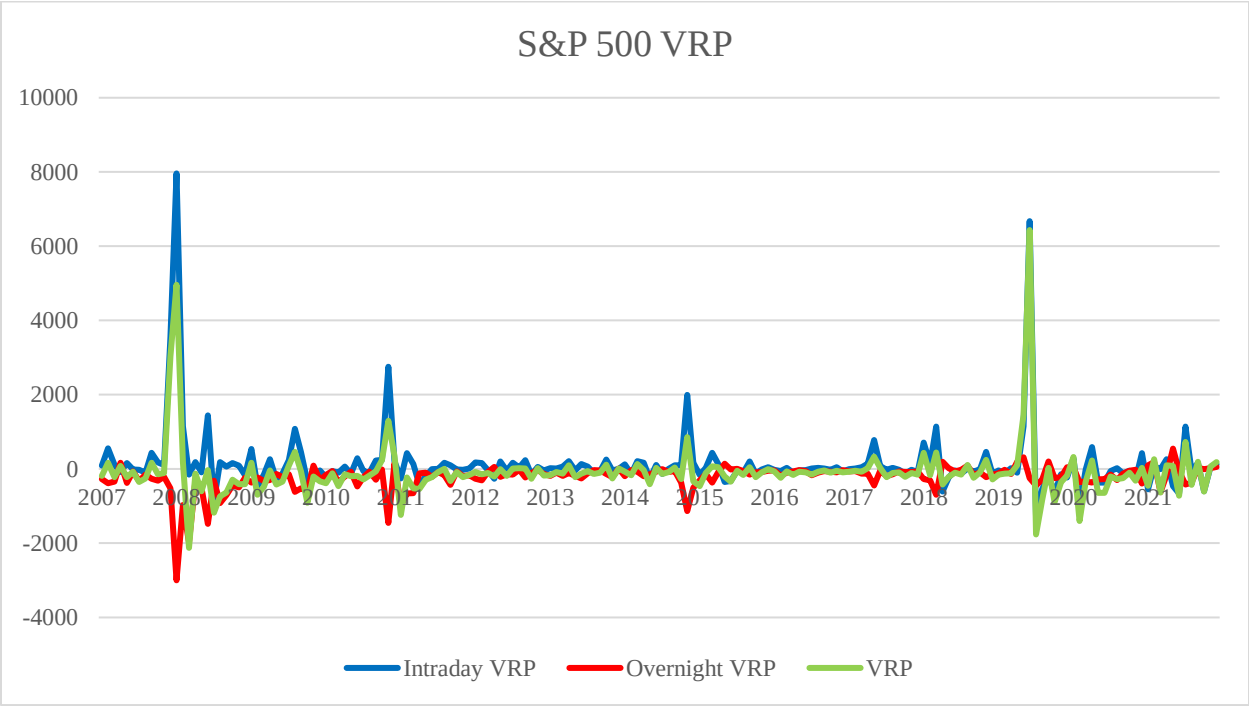
PANEL C: Open-to-Close

	1 month			3 months			6 months			9 months			1 Year		
	Constant	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**
SP500	0,00686**	-0,00002***	0,0614	0,01869**	-0,00002***	0,0432	0,03675**	-0,00002*	0,0108	0,05484**	-0,00001	-0,0023	0,07563***	-0,00001	-0,0048
	(2,2778)	(-4,2572)		(2,0706)	(-2,9145)		(2,1688)	(-1,7118)		(2,3859)	(-1,2568)		(2,8570)	(-0,6498)	
Nasdaq100	0,01069***	-0,00002***	0,0485	0,03058***	-0,00003***	0,0374	0,06133***	-0,00001	-0,0024	0,09345***	0,00000	-0,0056	0,12894***	0,00001	-0,0050
	(2,9050)	(-2,7816)		(2,7873)	(-3,3699)		(3,0774)	(-0,8793)		(3,5030)	(0,1329)		(4,1739)	(0,6208)	
Russell2000	0,00769**	-0,00001***	0,0475	0,01869	-0,00002**	0,0305	0,03182	-0,00001	-0,0045	0,04360	0,00001	-0,0025	0,05914*	0,00001	-0,0011
	(1,9875)	(-3,0235)		(1,5964)	(-2,2261)		(1,5150)	(-0,3549)		(1,5910)	(0,7476)		(1,8883)	(0,9824)	
SMI	0,00075	0,00000	-0,0050	0,00175	-0,00002	0,0044	0,00699	-0,00003	0,0090	0,01365	-0,00003	0,0076	0,02216	-0,00003	0,0007
	(0,2200)	(-0,2817)		(0,2039)	(-1,2072)		(0,4847)	(-1,4548)		(0,7612)	(-1,6415)		(1,0457)	(-1,1747)	
DAX	0,00362	-0,00001	0,0020	0,01060	-0,00001	0,0020	0,02235	0,00001	-0,0049	0,03695	0,00000	-0,0056	0,05126*	0,00002	-0,0033
	(0,9311)	(-1,3834)		(0,9815)	(-0,8998)		(1,1786)	(0,3445)		(1,5306)	(0,1055)		(1,8929)	(0,7467)	
EUROSTOXX50	0,00143	-0,00001*	0,0084	0,00231	-0,00001	0,0034	0,00504	-0,00001	-0,0046	0,00786	0,00000	-0,0056	0,01091	0,00002	-0,0018
	(0,3935)	(-1,8333)		(0,2118)	(-1,1161)		(0,2533)	(-0,5129)		(0,3046)	(0,0605)		(0,3753)	(1,3918)	
NIKKEI225	0,00816*	0,00000	-0,0079	0,02397**	0,00000	-0,0079	0,05076**	0,00000	-0,0078	0,07912***	0,00002	-0,0052	0,10579***	0,00002	-0,0044
	(1,8929)	(-0,0206)		(2,0019)	(0,0258)		(2,5547)	(0,1407)		(3,0379)	(0,6044)		(3,5518)	(0,7873)	
KOSPI	0,00081	0,00000	-0,0075	0,00246	-0,00003	0,0006	0,00944	0,00002	-0,0068	0,01429	-0,00005	0,0002	0,02017	-0,00006	0,0003
	(0,2079)	(-0,2133)		(0,2316)	(-0,9548)		(0,5032)	(0,3454)		(0,5722)	(-0,6485)		(0,6647)	(-0,7492)	
HSI	-0,00172	-0,00002	0,0021	-0,00452	0,00000	-0,0079	0,00262	-0,00008*	0,0236	0,00071	-0,00008*	0,0141	-0,00672	-0,00004	-0,0041
	(-0,3584)	(-1,0960)		(-0,3640)	(0,1046)		(0,1409)	(-1,7557)		(0,0296)	(-1,8836)		(-0,2306)	(-0,7519)	

Table 8: Predictive regressions using monthly total (close-to-close), overnight (close-to-open) and intraday (open-to-close) variance risk premiums. The symbols *, **, *** represent 10%, 5%, and 1% significance levels. The time period is from October 2007 until September 2022 (unbalanced panel).

	1 month			**3 months**			**6 months**			**9 months**			**1 Year**		
	Constant	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**	**Constant**	**VS (Lag)**	**Adj R^2**
Close -to- Close	0,00082	-0,00001***	0,0109	0,00566**	-0,00002***	0,0167	0,01554***	-0,00002***	0,0239	0,02719***	-0,00003***	0,0351	0,04309***	-0,00003***	0,0376
	(0,5525)	(-3,8774)		(2,1936)	(-3,6794)		(4,1130)	(-3,4417)		(5,8528)	(-3,8456)		(8,0677)	(-2,7680)	
Open -to- Close	0,00275*	-0,00001***	0,0078	0,00891***	-0,00002***	0,0155	0,01912***	-0,00001	0,0150	0,03184***	-0,00001	0,0226	0,04636***	0,00000	0,0306
	(1,8883)	(-3,3663)		(3,5176)	(-3,4797)		(5,1264)	(-1,3745)		(6,9445)	(-0,8732)		(8,8091)	(0,1347)	
Close -to- Open	0,00207	-0,00000	-0,0021	0,00821***	0,00000	0,0050	0,01373***	-0,00002**	0,0191	0,02250***	-0,00005***	0,0350	0,03603***	-0,00005***	0,0445
	(1,2732)	(-0,0997)		(2,9005)	(0,3269)		(3,3206)	(-2,5393)		(4,4355)	(-3,8335)		(6,1843)	(-3,9154)	

Figure 1
The S&P 500 monthly total variance risk premium, intraday variance risk premium and overnight variance risk premium for the time period October 2007 to September 2022.



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